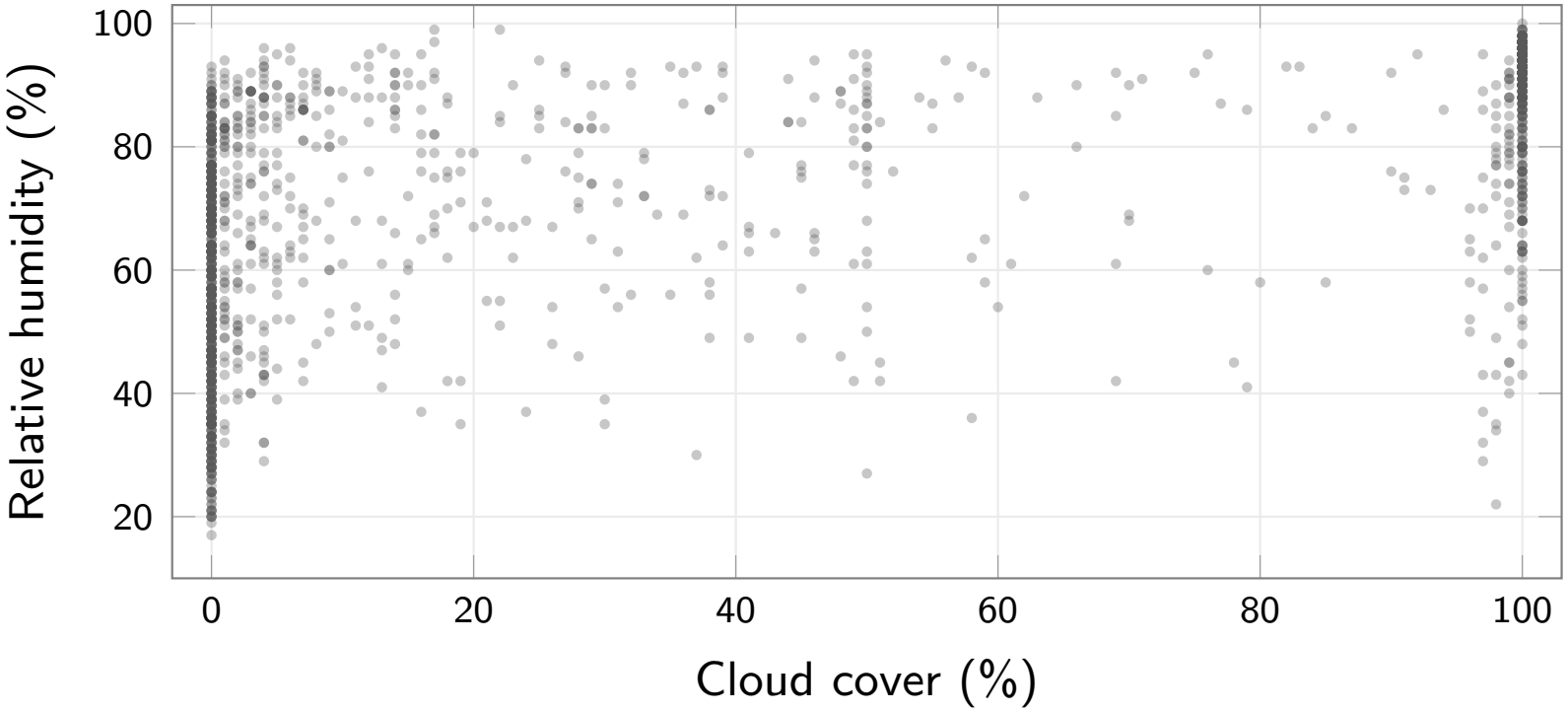
Unsupervised Learning: K-Means Clustering

Discovering patterns in Charlottesville weather

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DS 6021: Introduction to Predictive Modeling
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Can we discover types of local weather?



Each point is one weather hour. Where would you place the groups?

1,440 hourly model estimates; September 11–November 9, 2026 (UTC). Weather data: Open-Meteo.com, CC BY 4.0; reformatted and clustered for teaching.

What is being clustered?

- **One row:** one hour at a Charlottesville-area grid point.
- **Source:** recent hourly weather-model estimates from Open-Meteo.
- **Snapshot:** 60 completed UTC days, 1,440 rows, no missing values in the selected features.

UTC hour	Cloud (%)	Humidity (%)	Precip. (mm)
2026-09-11 00:00	61	61	0.0
2026-09-11 01:00	28	71	0.0
2026-09-11 02:00	49	77	0.0
2026-09-11 03:00	14	83	0.0

What these data represent

Hourly model estimates, not direct station measurements. Precipitation is the preceding hour's total, including rain, showers, and snow water equivalent.

Open-Meteo API documentation; retrieved November 10, 2026. Stored times are UTC; the dashboard displays local time.

From predicting labels to discovering patterns

Supervised learning

Observe features X and an outcome Y.

Learn a rule for predicting Y from X.

Unsupervised learning

Observe features X.

Look for useful structure without supplying an outcome.

	KNN	KMeans
What does K count?	Nearby observations	Cluster centers
What is learned?	Predictions from neighbors	Groups and their representatives
What do both need?	Meaningful features, distances, and scaling	

Features define the question

Start in two dimensions

$\mathbf{x}_i = (\text{cloud cover},\ \text{relative humidity})$.
Which hours have similar cloud and moisture conditions?

Then add a third dimension

$\mathbf{x}_i = (\text{cloud cover},\ \text{relative humidity},\ \text{precipitation})$.
Does including wetness change the grouping?

- Weather codes are **not** used as training labels or features.
- Time and temperature remain available for later interpretation.
- $K = 3$ is an exploratory choice, not evidence of three true weather types.

Scaling changes what "close" means

In raw units, squared distance in three dimensions is

$$d^2 = (\Delta\text{cloud})^2 + (\Delta\text{humidity})^2 + (\Delta\text{precipitation})^2.$$

- A 10-point cloud difference contributes 100.
- A 1-mm precipitation difference contributes 1.

A starting choice: standardize each feature

$$z_{ij} = \frac{x_{ij} - \bar{x}_j}{s_j}, \qquad s_j = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_{ij} - \bar{x}_j)^2}.$$

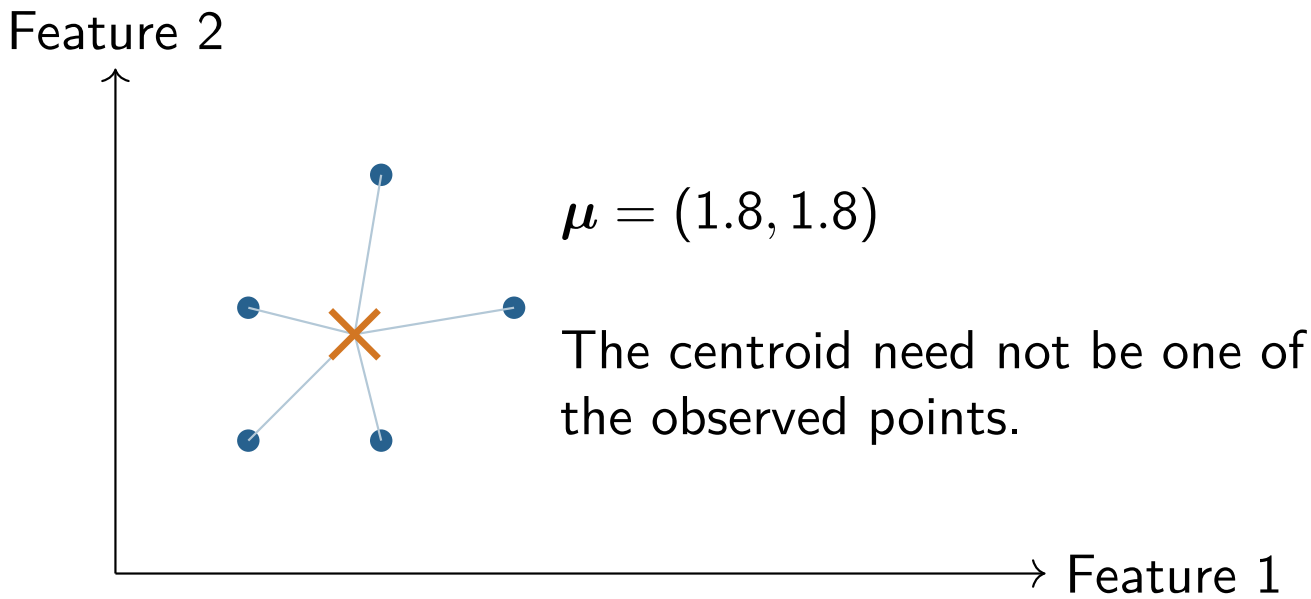
One unit is one feature standard deviation. This is the convention used by `StandardScaler`.

Scaling is a modeling choice. Rare large rainfall values can still matter greatly after standardization.

Represent each group by a center

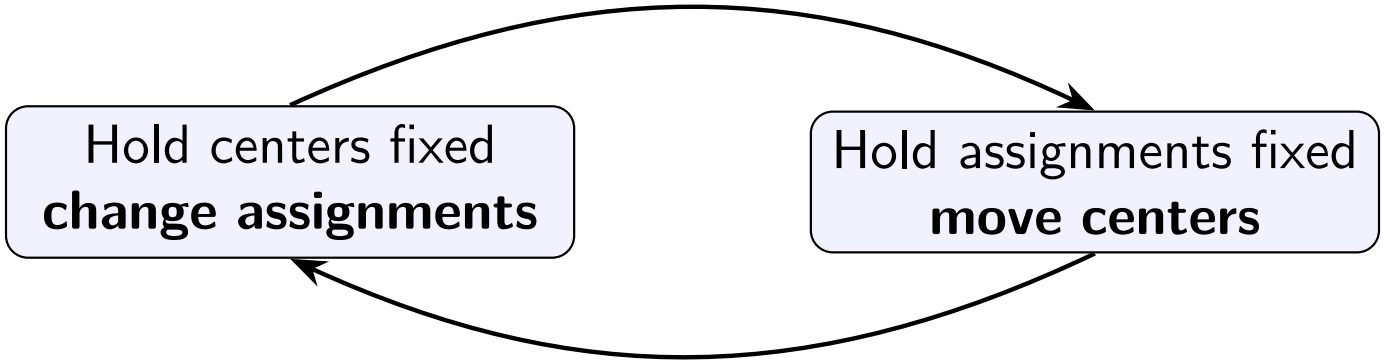
For cluster C_k, the centroid is the coordinatewise mean:

$$\boldsymbol{\mu}_k = \frac{1}{|C_k|} \sum_{i \in C_k} \mathbf{x}_i.$$



Lloyd's algorithm: assign, average, repeat

1. **Initialize:** choose K starting centers.
2. **Assign:** put each point in the nearest-center group.
3. **Update:** replace each center with its group's mean.
4. Repeat assignment and update until the grouping stabilizes.



Think about it

Why might moving a center change which points are closest to it?

The objective: total squared distance

Let n be the number of observations, p the number of features, and K the number of groups. Let $c_i \in \{1, \ldots, K\}$ assign point i to a group.

$$\min_{c_1,\ldots,c_n,\, \boldsymbol{\mu}_1,\ldots,\boldsymbol{\mu}_K} J = \sum_{i=1}^{n} \left\| \mathbf{x}_i - \boldsymbol{\mu}_{c_i} \right\|^2 = \sum_{k=1}^{K} \sum_{i \in C_k} \sum_{j=1}^{p} (x_{ij} - \mu_{kj})^2.$$

- J is the **within-cluster sum of squares** (WCSS), also called **inertia**.
- Every point contributes its squared distance to its assigned center.
- Regression SSE measured prediction errors; here SSE measures discrepancies from group representatives.

ISLP, Section 12.4.1; scikit-learn clustering guide.

Assignment: choose the nearest center

For fixed centers, assign point i by

$$c_i = \underset{k \in \{1,\ldots,K\}}{\arg\min} \left\| \mathbf{x}_i - \boldsymbol{\mu}_k \right\|^2 .$$

Example

For $\mathbf{x} = (3, 1)$ and centers $(1, 1.5)$ and $(6, 1.5)$:

$$d_1^2 = (3-1)^2 + (1-1.5)^2 = 4.25,$$
$$d_2^2 = (3-6)^2 + (1-1.5)^2 = 9.25.$$

Assign the point to group 1.

- Squared and ordinary Euclidean distance give the same nearest center.
- Resolve exact ties consistently, for example by the lowest cluster index.

Update: replace centers with group means

Hold assignments fixed. For each nonempty group,

$$\boldsymbol{\mu}_k^{\text{new}} = \frac{1}{|C_k|} \sum_{i \in C_k} \mathbf{x}_i.$$

Example

If one group contains $(1,1)$, $(1,2)$, and $(3,1)$, then

$$\boldsymbol{\mu}_1^{\text{new}} = \left(\frac{1+1+3}{3}, \frac{1+2+1}{3} \right) = \left(\frac{5}{3}, \frac{4}{3} \right).$$

Think about it

The center moves. Do the points move? Do their memberships change during this update?

Points stay fixed; memberships wait until the next assignment step.

Why does the mean minimize squared error?

Fix a group C_k and one feature j. Minimize

$$Q(\mu_{kj}) = \sum_{i \in C_k} (x_{ij} - \mu_{kj})^2.$$

$$\begin{aligned} \frac{\partial Q}{\partial \mu_{kj}} &= -2 \sum_{i \in C_k} (x_{ij} - \mu_{kj}) \\ &= -2 \sum_{i \in C_k} x_{ij} + 2|C_k| \mu_{kj} = 0. \end{aligned}$$

Therefore,

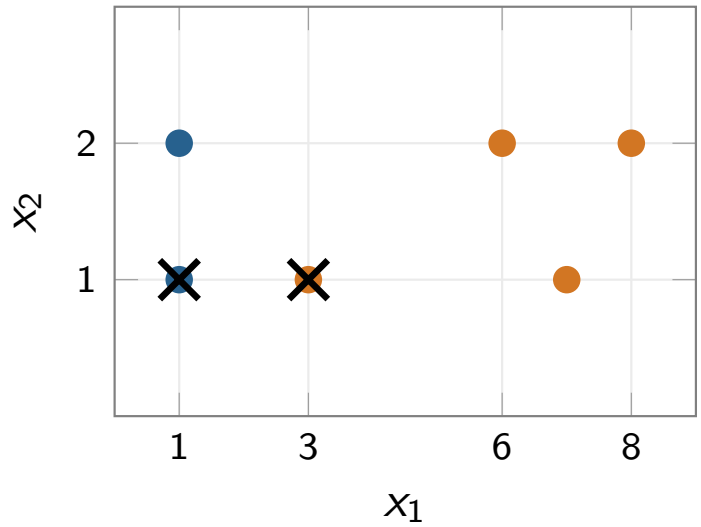
$$\boxed{\widehat{\mu}_{kj} = \frac{1}{|C_k|} \sum_{i \in C_k} x_{ij}}.$$

The second derivative is $2|C_k| > 0$, so this is a minimum. Apply the result to every feature.

A six-point example: first assignment

Use $K = 2$ with initial centers $\boldsymbol{\mu}_1 = (1, 1)$ and $\boldsymbol{\mu}_2 = (3, 1)$.

Point	d_1^2	d_2^2	Group
A $(1, 1)$	0	4	1
B $(1, 2)$	1	5	1
C $(3, 1)$	4	0	2
D $(6, 2)$	26	10	2
E $(7, 1)$	36	16	2
F $(8, 2)$	50	26	2

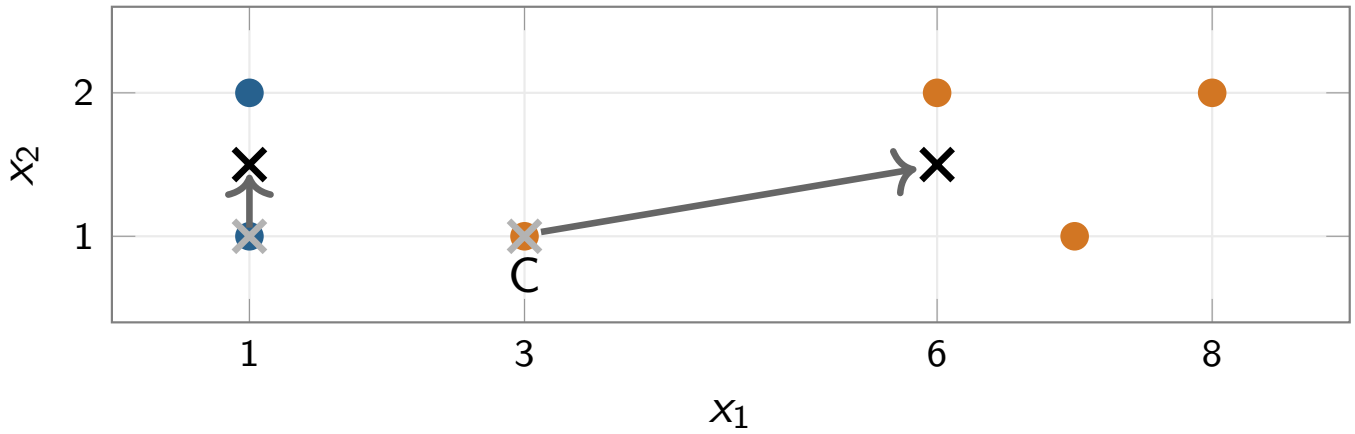


$$J = 0 + 1 + 0 + 10 + 16 + 26 = 53.$$

Illustrative points in arbitrary numerical units; not sampled weather data.

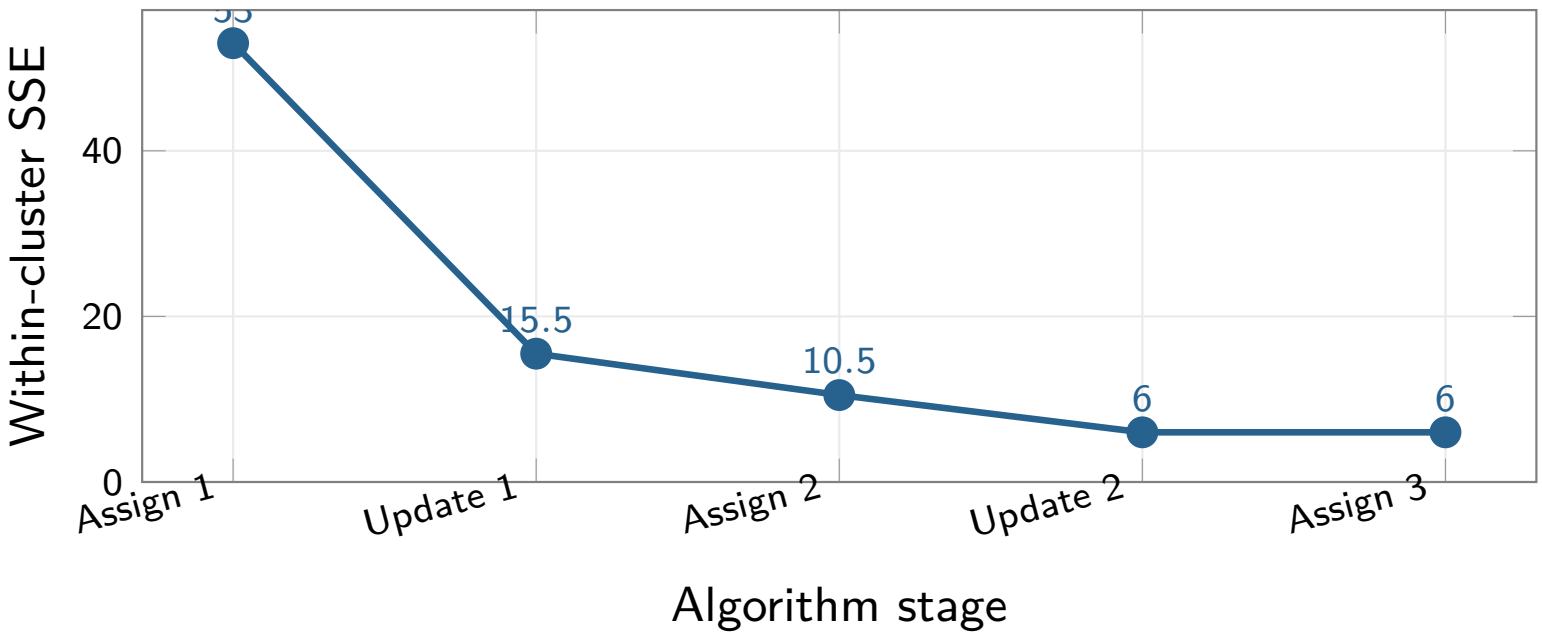
Move the centers, then reconsider membership

First update: $\boldsymbol{\mu}_1 = (1, 1.5)$ and $\boldsymbol{\mu}_2 = (6, 1.5)$.



- Update: keep groups fixed; SSE falls from 53 to 15.5.
- Reassign: C is now closer to center 1; SSE falls to 10.5.
- Update again: centers become $(5/3, 4/3)$ and $(7, 5/3)$; SSE is 6.

The objective falls and then levels off



Converged grouping

Group 1: A, B, C. Group 2: D, E, F.

Another assignment makes no changes. The objective can stay equal.

Why can't an exact step increase the objective?

1. **Assignment:** each point chooses its closest fixed center. Its squared-distance contribution cannot get larger.
2. **Update:** the mean minimizes squared distance for each fixed group. Each group's contribution cannot get larger.

$$J_{\text{after assignment}} \leq J_{\text{before}}, \qquad J_{\text{after update}} \leq J_{\text{after assignment}}.$$

What convergence means

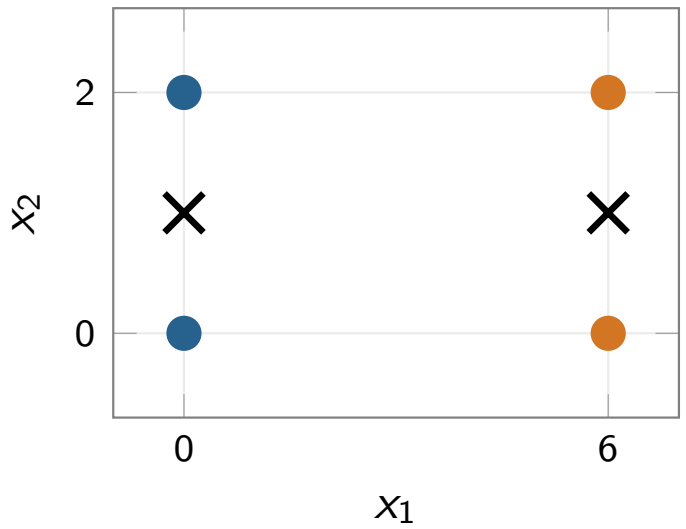
With consistent ties and valid updates, the ideal finite-data algorithm settles at a fixed grouping. Implementations may also stop at a tolerance or iteration limit.

A stable solution need not be the global minimum.

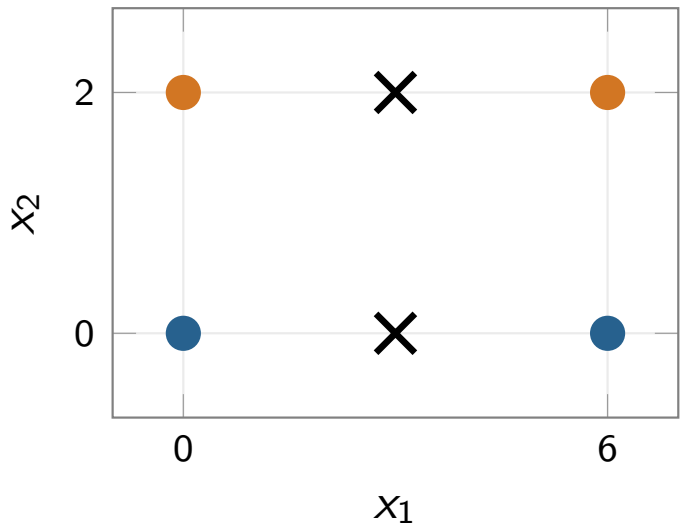
Convergence does not guarantee the best grouping

Same four observations, same $K = 2$, two stable solutions:

Split left and right: SSE $= 4$



Split bottom and top: SSE $= 36$



Think about it

In the right panel, why does neither alternating step change anything?

Illustrative points; crosses are centroids. Different initializations can reach different solutions.

Better starting centers and multiple restarts

KMeans++ initialization

Choose the first center randomly. Favor later points that are far from all chosen centers:

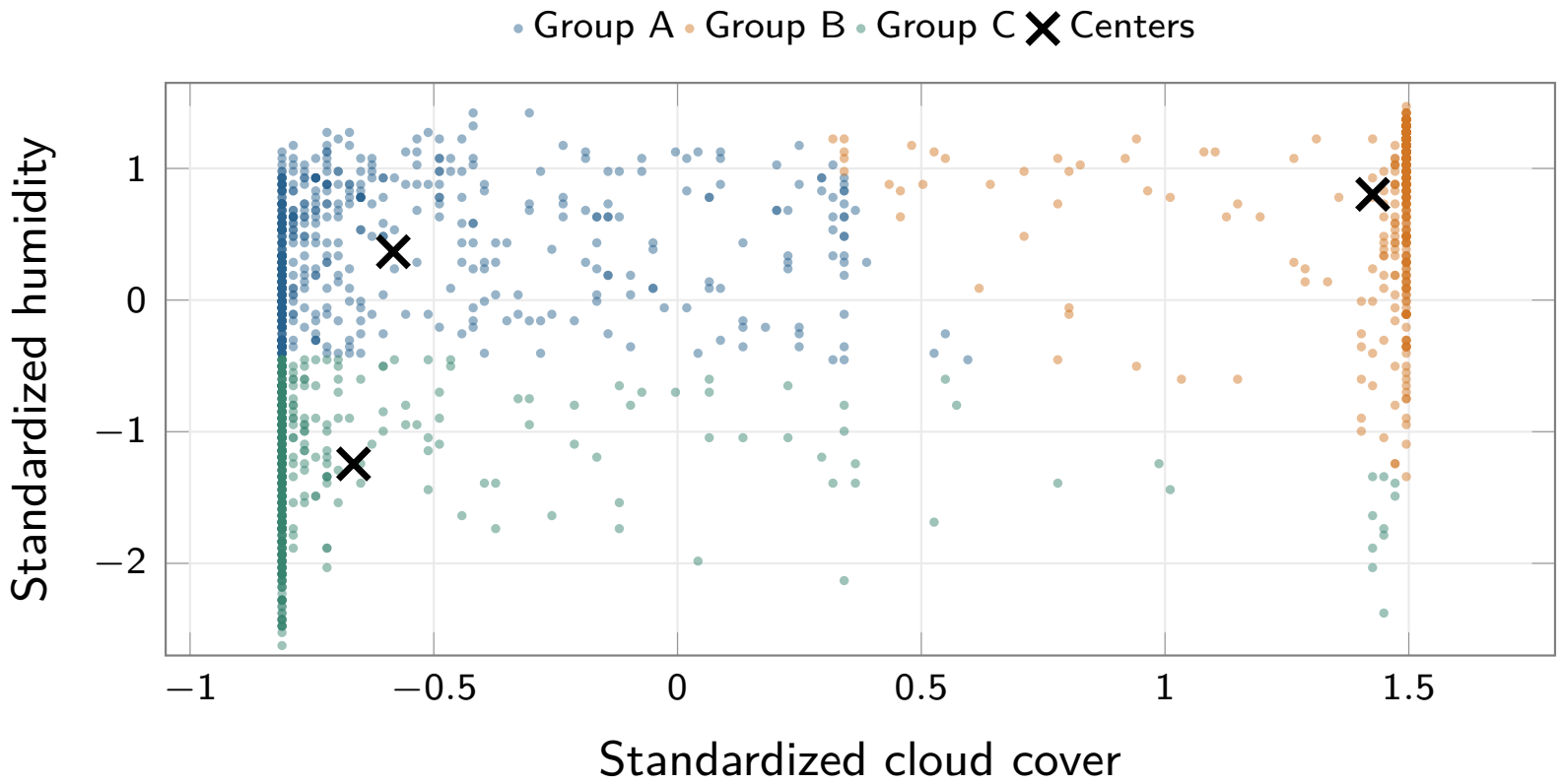
$$P(\mathbf{x}_i \text{ chosen next}) \propto D(\mathbf{x}_i)^2,$$

where $D(\mathbf{x}_i)$ is distance to its nearest chosen center.

- Run the full algorithm from multiple starts.
- Keep the run with the lowest final inertia for the **same data, features, scaling, and** K.
- Use a fixed random seed to reproduce the experiment.
- The notebook uses 20 starts; better initialization does not guarantee a global optimum.

scikit-learn KMeans API; its implementation uses a greedy variant of KMeans++.

The weather data in two dimensions



$K = 3$: groups contain 558, 436, and 446 hours. The geometry uses standardized features; the points themselves have not changed.

All 1,440 snapshot hours; KMeans++, 20 starts, seed 6021, tolerance 0. Weather data: Open-Meteo.com, CC BY 4.0; reformatted and clustered for teaching.

Dashboard: watch one run converge

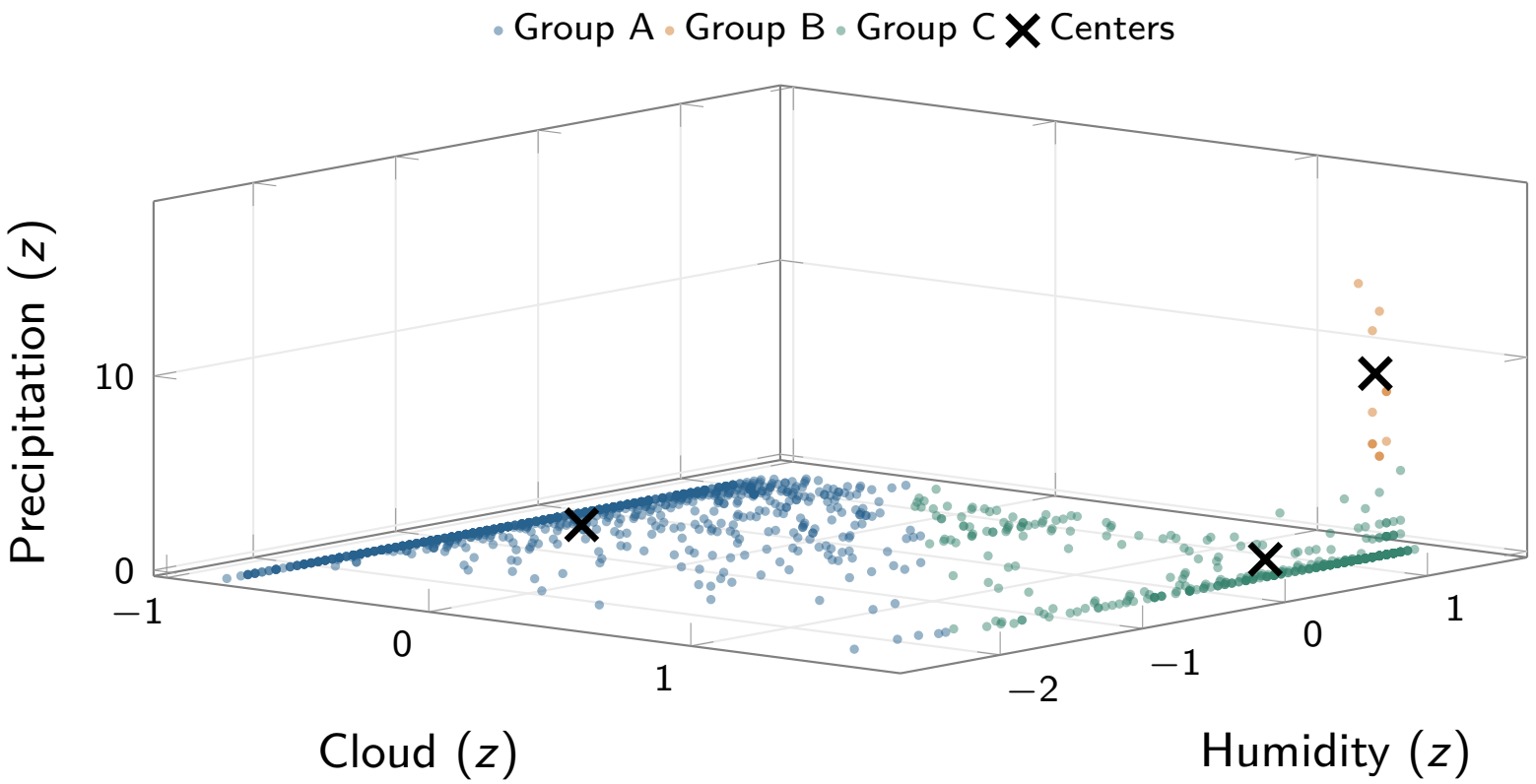
1. Start with the saved weather snapshot, **2D**, $K = 3$, and standardization enabled.
2. Step through **assignment**. Which observations change color?
3. Step through **update**. Which centers move, and why?
4. Follow the centroid paths and within-cluster SSE.
5. Change the seed and compare the final result. Then add precipitation in **3D**.

Prediction before clicking

Which quantities are allowed to change in the next step?
Will the objective increase, decrease, or stay the same?

Open the KMeans weather dashboard

Adding precipitation changes the geometry



The same 1,440 hours are refitted in **three features**. A small group of wetter hours now uses one center.

All features standardized; crosses show centers. Group letters are arbitrary within this fit. Weather data: Open-Meteo.com, CC BY 4.0; reformatted and clustered for teaching.

Interpret centers in the original units

Two-feature fit: cloud cover and relative humidity.

Group	Hours	Mean cloud (%)	Mean humidity (%)
A	558	9.9	77.6
B	436	97.0	86.5
C	446	6.4	45.0

Three-feature fit: add precipitation.

Group	Hours	Cloud (%)	Humidity (%)	Precip. (mm)
A	954	6.2	62.1	0.00
B	12	100.0	94.4	4.04
C	474	91.8	85.8	0.04

These are group means. Letters identify groups within a fit; they do not identify the same group across fits.

Reference fits: KMeans++, 20 starts, seed 6021, Lloyd algorithm, tolerance 0. Values rounded.

Do these groups mean sunny, cloudy, and rainy?

- The 2D fit separates cloudiness and humidity, but does not directly use precipitation.
- Only **44 of 1,440 hours** have positive precipitation in this snapshot.
- In the 3D fit, 12 wetter hours form one group; another group includes 32 lightly wet hours and many dry hours.
- All hours are included. Low cloud cover means **clearer**, not necessarily sunny at night.

Interpret after fitting

Use names such as "clearer and drier" or "very wet and overcast" as descriptions of the profiles. They are not established weather classes.

How many clusters should we request?

- A small K gives a simpler summary but may merge different patterns.
- A large K gives finer subdivisions and lower attainable SSE.
- At $K = n$, each point can be its own center and SSE is zero.

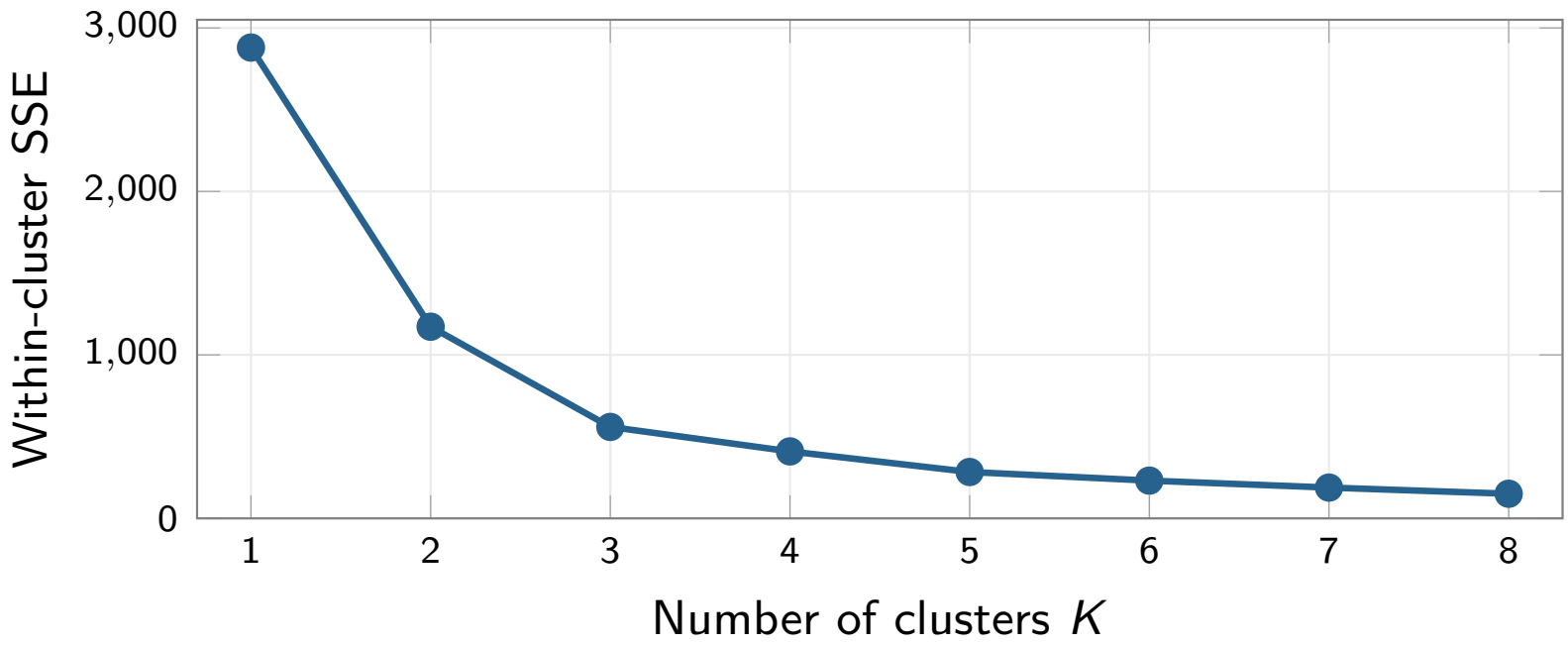
Three kinds of evidence

1. **Compactness:** how much does another center reduce SSE?
2. **Separation:** how well do points fit their own group?
3. **Usefulness and stability:** are profiles meaningful and reproducible?

Think about it

Why is choosing the K with the smallest SSE insufficient?

The elbow: diminishing returns from more centers



Here $K = 3$ is a reasonable compact summary; $K = 4$ refines it. An elbow is a judgment, not an automatic proof of the true number of groups.

Same 2D standardized weather data for every K; best of 20 starts, seed 6021.

Silhouette: cohesion compared with separation

For point i:

- $a(i)$: average distance to the **other points** in its own group.
- $b(i)$: smallest average distance to the points in any other group.

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}, \qquad -1 \leq s(i) \leq 1.$$

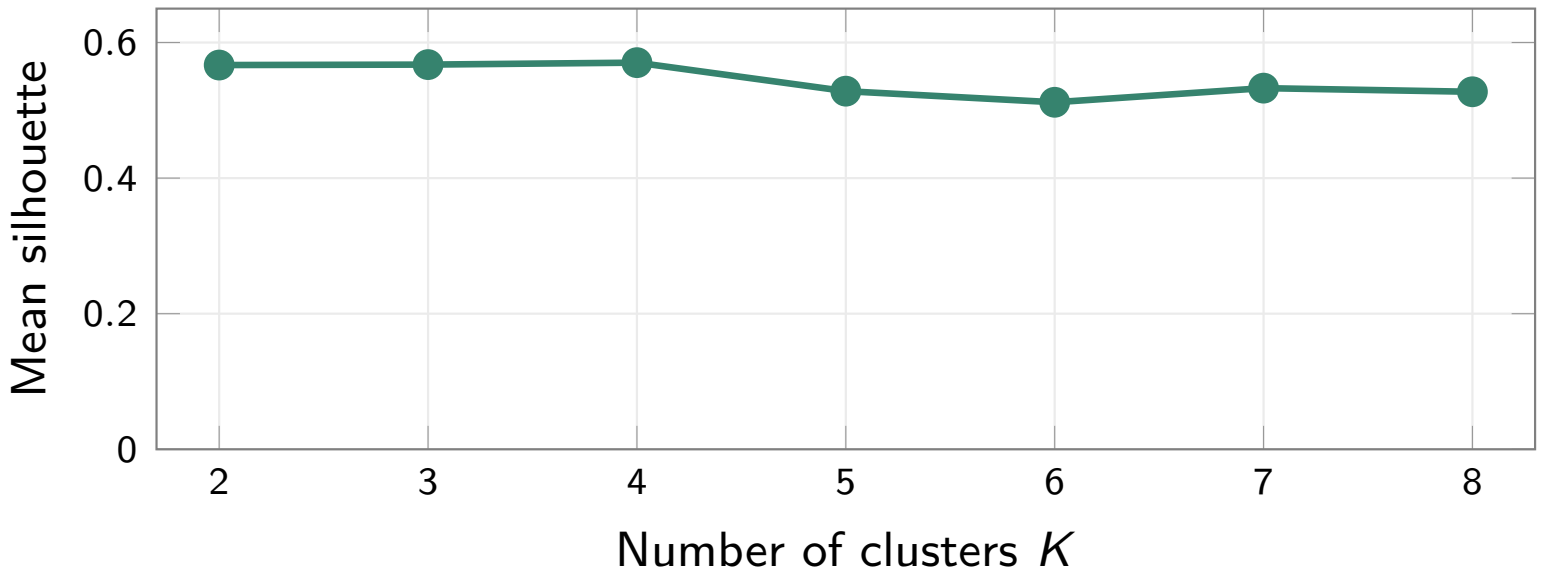
- Near 1: relatively cohesive and well separated.
- Near 0: groups overlap around this point.
- Negative: another group is closer on average.

Example

If $a(i) = 2$ and $b(i) = 5$, then $s(i) = 3/5 = 0.6$.

Use ordinary Euclidean distances in the same standardized space. Silhouette documentation.

Read diagnostics together, then inspect the groups



In this 2D snapshot, scores for $K = 2, 3, 4$ are similar: approximately 0.567, 0.567, and 0.570. A tiny improvement does not settle the substantive choice.

- Compare profiles and sizes; repeat across seeds and date windows.
- A higher silhouette does not establish meteorological truth.

When should we question the result?

- **Outliers:** squared distance gives extreme values large influence.
- **Geometry:** elongated, curved, or differently dispersed groups may not fit nearest-center partitions well.
- **Representation:** redundant features can count similar information twice.
- **Weather sampling:** many dry hours, day/night effects, seasons, and adjacent-hour dependence affect what we see.
- **Live refresh:** a new window changes the dataset, and potentially the clusters.

A defensible analysis

Explain the feature and scaling choices, inspect the profiles, examine stability, and report limitations with the grouping.

Explain, interpret, and diagnose

1. Why does the assignment step hold centers fixed?
2. Why does the update use a mean rather than an arbitrary point?
3. Can SSE stop changing even though another initialization gives a better result?
4. Why can adding precipitation change the cloud/humidity grouping?
5. What evidence would justify naming a cluster “wet”? What would the name leave out?

Connection to modern data science

The same reasoning applies to grouping customers, documents, sensor states, or learned embeddings: representation determines similarity, and clusters require interpretation.

Readings and companion materials

Core reading

- James, Witten, Hastie, Tibshirani, and Taylor, *An Introduction to Statistical Learning with Applications in Python*: Sections 12.1, 12.4.1, and 12.4.3. statlearning.com

Computation and interpretation

- Companion Colab notebook: inspect the data, fit KMeans, visualize groups, and compare K and initialization.
- scikit-learn clustering guide and KMeans API reference.
- Open-Meteo weather variables and API documentation.

Optional deeper reading

- Hastie, Tibshirani, and Friedman, *The Elements of Statistical Learning*, Section 14.3.6 (K-means). hastie.su.domains/ElemStatLearn

Optional: the gap statistic

Does the clustering improve on what we would expect from a reference dataset with no comparable group structure?

$$\mathrm{Gap}(K) = \mathbb{E}^*[\log W_K^*] - \log W_K,$$

where W_K is the observed within-cluster dispersion and W_K^* comes from reference datasets.

- Simulate reference datasets, often from a suitable uniform box.
- Cluster each with the same method to estimate the expected log dispersion.

Original selection rule

Choose the smallest K such that

$$\mathrm{Gap}(K) \geq \mathrm{Gap}(K+1) - s_{K+1},$$

where s_{K+1} includes simulation uncertainty.

Tibshirani, Walther, and Hastie (2001). Estimating the number of clusters in a data set via the gap statistic.

Optional: ties, empty groups, and stopping

- **Tie:** assign to one center using a fixed convention.
- **Empty group:** its mean is undefined. Keep the old center or reinitialize according to an explicit policy.
- **Identical observations:** there cannot be more distinct positive-size geometric groups than distinct feature vectors.
- **Stopping:** stable assignments, sufficiently small movement, or an iteration limit are different stopping criteria.

For the reference tables in these notes

We use zero tolerance ($\texttt{tol} = 0$), at most 300 iterations per start, 20 starts, and seed 6021. The reported centers equal their final group means up to numerical precision.

Optional: categorical features and Gower distance

- Standard KMeans uses numerical coordinates, squared Euclidean distance, and arithmetic means.
- A nominal label such as "cloudy" does not have a meaningful arithmetic mean.
- Integer coding can create arbitrary distances. One-hot encoding changes the geometry and feature weighting.
- Gower distance can combine appropriately scaled numerical and categorical comparisons, but it is not the standard KMeans objective.

Connect to KNN

KNN can use a chosen pairwise distance directly. KMeans also needs a compatible center and update rule. For general dissimilarities, consider methods such as K-medoids or suitable hierarchical clustering.

Optional: choosing another clustering approach

Method	Useful perspective	A question to ask
KMeans	Summarize with K numerical means	Are compact nearest-center groups useful?
Hierarchical	Build nested groupings with a linkage rule	Which distance and linkage express similarity?
Density based	Seek connected dense regions; may mark noise	What density scale and neighborhood are appropriate?
K-medoids	Represent groups by observed points	Is a general dissimilarity or an observed representative needed?

Hierarchical behavior depends on linkage: Ward favors compact groups; single linkage can follow irregular shapes but may chain.

scikit-learn clustering guide; ISLP, Section 12.4.2 (optional next topic).